Experiment No. 10

Crack FTP Credentials Using a Dictionary Attack

AIM

To demonstrate the weakness of using default or weak FTP credentials by performing a dictionary attack against an FTP service in a controlled lab environment using a tool such as Hydra.

PROCEDURE

1. Set up a vulnerable FTP server (e.g., vsftpd) in an isolated lab environment with a known test user account configured with a weak password.
2. Prepare/choose a username wordlist (users.txt) and a password dictionary wordlist (passwords.txt) for the attack.
3. Verify the FTP service is reachable using an FTP client or by checking port 21 with Nmap (nmap -p21).
4. Launch the THC-Hydra tool with the FTP module, specifying the target IP, username list, and password dictionary: hydra -L users.txt -P passwords.txt ftp://.
5. Allow Hydra to systematically attempt each username/password combination against the FTP login service.
6. Observe Hydra's output for any successful login combination reported.
7. Verify the discovered credentials by manually logging into the FTP server, and record the result.

SAMPLE OUTPUT

```
$ hydra -L users.txt -P passwords.txt ftp://192.168.1.50
Hydra v9.4 starting at 2026-11-01 10:12:03
[DATA] attacking ftp://192.168.1.50:21/
[21][ftp] host: 192.168.1.50   login: testuser   password: admin123
1 of 1 target successfully completed, 1 valid password found

Manual verification:
$ ftp 192.168.1.50
Name: testuser   Password: admin123
230 Login successful.
```

RESULT

A dictionary attack against the FTP service was successfully carried out using Hydra, recovering the weak credentials (testuser / admin123), demonstrating the importance of strong password policies and account lockout mechanisms for FTP services.